The Weight Between and the Exit Line

Cross-block bounds for the boundary census, and where opened matter leaves the disk

Jeromie Beasley

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Abstract

Cut a fermionic system into a region A and its complement B. The one-body correlation matrix C then has a cross block, and $X_A = C_{AB}C_{BA}$ is the weight between the two sides. We prove that X_A is positive and bounded by $\frac{1}{4}$. It is two-sided: X_A and X_B share their positive spectrum. It is stable in operator norm with a certified margin. It lies below the local defect $C_A - C_A^2$. For a pure Gaussian (Slater) state, C is an orthogonal projector and $X_A = A_C - A_C^2$. Four times its trace bounds the Gaussian mutual information from below, with a sharp constant and no upper constant without a spectral margin. The signed boundary offset $\tau = \log\det(I - C_A)/\det C_A$ moves no faster than $\dot{\tau}^2 \le 4gS$. Two preparations with the same census, offset, cross weight and coupling saturate this bound in opposite directions.

On the PT-symmetric Su–Schrieffer–Heeger chain the correlation matrix is a biorthogonal idempotent. There the radial census survives and positivity does not. The band-averaged Petermann factor equals 2 exactly on the line $E(0)E(\pi) = u^2$, equivalently $(w^2-v^2)^2 = 2u^2(w^2+v^2)$.

We construct a linearly weighted operator $T_R = \frac{1}{2}\{X + \frac{1}{2}, H\}_R$ that commutes entrywise with the half-chain correlation block. The proof uses only $H(k)^2 = E(k)^2 I$, so it holds without Hermiticity. At 200 sampled gapped parameter points, its tested low-index edge eigenvectors exhibit an equally spaced entanglement ladder $\varepsilon_j = j\Delta + i\theta$ with a shared phase described by an elliptic integral. The spacing is $\Delta = 2\pi K'(k)/K(k)$, with modulus $k = v'/w'$ taken from the Hermitian band hidden in the opened one. The phase is $\theta = \pi F(\arcsin s \,|\, k)/K(k)$, with $s = uw/(w^2 - v'^2)$. On the tested levels the eigenvector residual is below 8×10^{-12}, the spacing agrees to 2×10^{-8}, the phase to 4×10^{-7}, and the inside/outside verdict holds at 200 of 200 points. For this proposed ladder, we prove simultaneous crossing of the Cayley unit circle and show that its phase reaches $\pi/2$ exactly on the band-averaged Petermann line $\langle K \rangle = 2$, in both directions, the converse through an explicit factorisation. The dimer limit is proved exactly. An analytic derivation of the ladder for the coupled chain remains open.

Grades. [T] proved here; [L] also checked in Lean in the author's formal suite; [V] verified numerically with the stated controls; [R] a known result in this coordinate; [C] open.

1 Introduction

Every finite record of a cut has two readings. One is unsigned: it counts how many boundary levels are partially occupied. The other is signed: it says which way each level leans. Both readings live

on one chart,

$$\Gamma = \frac{z-1}{z+1} = \tanh\frac{\varepsilon}{2} = 1-2\nu, \qquad z = \frac{1-\nu}{\nu} = e^{\varepsilon},$$

for an occupation ν of the region's correlation matrix. The census is the gauge without its sign. The offset is the clock face. This paper proves two things about this chart.

The first is about the *weight between* the two sides of a cut, the cross block $X_A = C_{AB}C_{BA}$. Section 3 proves its order, size, symmetry, stability and information bounds. It also gives exact examples marking where the one-body record stops speaking, and proves the speed limit of the signed offset.

The second is about where *opened* matter leaves the disk. Opening means a non-Hermitian generator. On the PT-symmetric dimerized chain the levels leave $|\Gamma| < 1$ inside the gapped phase, before the critical point. Measured on finite blocks, that exit sits on the line $E(0)E(\pi) = u^2$, where the band-averaged Petermann factor equals 2 (Section 4), and the question is why it happens there. Section 5 gives the mechanism. An operator commutes exactly with the half-chain block, and numerically, the tested edge levels follow a ladder with one shared phase. A ladder with one phase crosses the rim all at once, and its phase reaches a quarter turn exactly on the Petermann line. The exact commutation, the simultaneous crossing, the location of the crossing and a closed dimer case are proved. The ladder form and the elliptic phase formula are measured at 200 random points. Deriving them analytically is the remaining step.

2 The chart

Proposition 2.1 (the bridge)**.** $\Gamma = \tanh(\varepsilon/2) = 1-2\nu$*, and* $\nu(1-\nu) = (1-\Gamma^2)/4$*. For real* ν *the window* $\nu \in [\delta, 1-\delta]$ *maps to the real diameter segment* $[-1+2\delta, 1-2\delta]$*, equivalently* $|\Gamma| \le 1-2\delta$ *for real* ν*. [T][L]*

Proof. $z-1 = (1-2\nu)/\nu$ and $z+1 = 1/\nu$; $(e^{\varepsilon}-1)/(e^{\varepsilon}+1) = \tanh(\varepsilon/2)$; $1-(1-2\nu)^2 = 4\nu(1-\nu)$. □

Summed over the levels of the region, the rapidity is the signed offset

$$\tau = \sum_{\nu} \varepsilon_{\nu} = \mathrm{Tr}[\log(I-C) - \log C] = \log\frac{\det(I-C)}{\det C}, \tag{1}$$

which for a number-conserving Gaussian state is $\log P(N_A = 0)/P(N_A = |A|)$. The reduced state is a product over the eigenmodes of C, so $P(0) = \prod(1-\nu)$ and $P(|A|) = \prod \nu$ [T]. The census counts levels with $\nu(1-\nu)$ above a threshold, and is blind to the sign of Γ. For example, $\mathrm{diag}(0.2, 0.8)$ and $\mathrm{diag}(0.2, 0.2)$ have the same defect $C - C^2 = 0.16I$ and offsets 0 and $\log 16$.

Theorem 2.2 (the Cayley domain)**.** *For* $z \neq -1$, $|z+1|^2 - |z-1|^2 = 4\,\mathrm{Re}\, z$*, so* $|\Gamma| < 1 \iff \mathrm{Re}\, z > 0$ *and* $|\Gamma| = 1 \iff \mathrm{Re}\, z = 0$*. For* $z = e^{W}$, $|\tanh(W/2)| < 1 \iff \cos(\mathrm{Im}\, W) > 0$*; on the principal branch this is* $|\mathrm{Im}\, W| < \pi/2$*. [T]*

Proof. Expand the moduli; $\mathrm{Re}\, e^{W} = e^{\mathrm{Re}\, W}\cos \mathrm{Im}\, W$. □

The real cases are checked in Lean: a positive ratio lands strictly inside, a negative ratio $S/(-s)$ with $S > s > 0$ lands outside at rim distance $2s/(S-s)$, and $z = -1$ is the pole. A complex ratio does not by itself leave the disk: $z = 1 - i\gamma$ has $|\Gamma|^2 = \gamma^2/(4+\gamma^2) < 1$. Leaving needs the real part of the ratio to change sign. In the rapidity this is a quarter turn of the phase.

3 The weight between

Let $C = C^*$ with $0 \le C \le I$ on $\mathcal{H}_A \oplus \mathcal{H}_B$, of dimensions a, b, and write

$$C = \begin{pmatrix} A_C & B_C \\ B_C^* & D_C \end{pmatrix}, \quad F_A = A_C - A_C^2, \quad T_A = (C - C^2)_{AA}, \quad X_A = B_C B_C^*, \quad X_B = B_C^* B_C.$$

Fix $0 < \varepsilon < \frac{1}{2}$ and put $\vartheta = \varepsilon(1-\varepsilon)$. The *raw census* is $N_\varepsilon^{\mathrm{raw}} = \operatorname{rank} \mathbf{1}_{[\varepsilon, 1-\varepsilon]}(A_C)$. The *boundary census* is $N_\varepsilon^{\partial} = \operatorname{rank} \mathbf{1}_{[\vartheta, \infty)}(X_A)$. Counts include multiplicity and equality at the threshold.

Theorem 3.1 (the weight between)**.** *For every such C:*

1. *(defect split) $F_A = T_A + X_A$; with $C^2 = C$, $X_A = A_C - A_C^2$, and this needs no Hermiticity. [T][L]*
2. *(order and size) $0 \le X_A \le F_A \le \frac{1}{4} I$, $0 \le T_A \le F_A$, $\|B_C\| \le \frac{1}{2}$, attained by $C = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. [T][L]*
3. *(two-sided) X_A and X_B have the same positive eigenvalues with multiplicity. So $N_\varepsilon^{\partial}(A) = N_\varepsilon^{\partial}(B) \le \operatorname{rank} B_C \le \min(a, b)$, and $\operatorname{Tr} X_A = \operatorname{Tr} X_B = \|B_C\|_F^2 \le \frac{1}{4}\min(a, b)$. [T]*
4. *(comparison and recovery) $N_\varepsilon^{\partial} \le N_\varepsilon^{\mathrm{raw}}$, with equality when $C^2 = C$. [T]*
5. *(trace) $\vartheta N_\varepsilon^{\partial} \le \operatorname{Tr} X_A = \sum_{i \in A, j \in B} |C_{ij}|^2$, and $\operatorname{Tr} X_A = 0 \iff B_C = 0$. [T][L]*
6. *(local unitaries) Conjugation by $U_A \oplus U_B$ leaves every census, spectrum and trace above unchanged. [T]*

Proof. (1) $(C^2)_{AA} = A_C^2 + B_C B_C^*$.

(2) Compression preserves order. $C - C^2 \ge 0$ by the spectral theorem, hence $T_A \ge 0$. Then (1) gives the two comparisons, and $F_A = \frac{1}{4} I - (A_C - \frac{1}{2})^2 \le \frac{1}{4} I$, so $\|B_C\|^2 = \|X_A\| \le \frac{1}{4}$.

(3) If $B_C B_C^* u = \lambda u$ with $\lambda > 0$, then $B_C^* u \ne 0$ is an eigenvector of $B_C^* B_C$ with eigenvalue λ. The inverse map is B_C/λ, so the multiplicities agree.

(4) $\nu(1-\nu) - \vartheta = (\nu - \varepsilon)(1 - \varepsilon - \nu)$, so $\operatorname{rank} \mathbf{1}_{[\vartheta, \infty)}(F_A) = N_\varepsilon^{\mathrm{raw}}$. By min–max, $\lambda_j(X_A) \le \lambda_j(F_A)$. If $C^2 = C$ then $T_A = 0$.

(5) Each counted eigenvalue contributes at least ϑ to the trace.

(6) Block multiplication. □

The boundary coordinate. Each eigenvalue $\lambda \in [0, \frac{1}{4}]$ of X_A has the radius $\Gamma_b = \sqrt{1 - 4\lambda} \in [0, 1]$. For a pure Gaussian (Slater) state, C is an orthogonal projector, $X_A = A_C - A_C^2$ and $\lambda = \nu(1-\nu)$, so $\Gamma_b = |1 - 2\nu|$: the boundary coordinate is the modulus of the census point and forgets its sign [T][L]. For mixed states X_A need not commute with A_C, and only the order comparisons remain.

Theorem 3.2 (stability with a certified count)**.** *Let $C, \tilde{C}$ be positive contractions on the same partition, with $\delta = \|C - \tilde{C}\|$. Then $\|X_A(C) - X_A(\tilde{C})\| \le \delta$. With $\mathcal{N}_Y(s) = \operatorname{rank} \mathbf{1}_{[s, \infty)}(Y)$, for $\delta < \vartheta$,*

$$\mathcal{N}_{X_A(C)}(\vartheta + \delta) \le N_\varepsilon^{\partial}(\tilde{C}) \le \mathcal{N}_{X_A(C)}(\vartheta - \delta).$$

If $\operatorname{dist}(\vartheta, \operatorname{spec} X_A(C)) > \delta$ the two counts are equal. [T]

Proof. $\|BB^* - \tilde{B}\tilde{B}^*\| \le (\|B\| + \|\tilde{B}\|)\|B - \tilde{B}\| \le \delta$ by Theorem 3.1(2). Then apply Weyl's inequality. □

Theorem 3.3 (near-projector comparison)**.** *With $\eta = \|C - C^2\|$, $X_A \le F_A \le X_A + \eta I$. Hence $\mathcal{N}_{F_A}(\vartheta + \eta) \le N_\varepsilon^\partial \le N_\varepsilon^{\mathrm{raw}}$. [T]*

Proof. $0 \le C - C^2 \le \eta I$ compresses to $0 \le T_A \le \eta I$. Then use (1) and ordered-eigenvalue monotonicity. □

Proposition 3.4 (the coordinate is Hölder)**.** $|\Gamma_b(\lambda) - \Gamma_b(\tilde{\lambda})| \le 2\sqrt{|\lambda - \tilde{\lambda}|}$. *If both radii are at least $g > 0$, the bound is $(2/g)|\lambda - \tilde{\lambda}|$. The eigenvalue step does not transfer to nonnormal matrices: $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ \delta & 0 \end{pmatrix}$ differ by δ, while their eigenvalues move by $\sqrt{\delta}$. [T]*

Proof. $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$; rationalise for the second bound. □

What the one-body record cannot say.

Proposition 3.5 (same local block, different weight between)**.** *Every positive contraction A has the projector extension $P_A = \begin{pmatrix} A & \sqrt{A - A^2} \\ \sqrt{A - A^2} & I - A \end{pmatrix}$, with $X_A(P_A) = A - A^2$. The disconnected $A \oplus (I - A)$ has the same local block and $X_A = 0$. [T]*

Proof. The square root commutes with A. Multiplying the blocks gives $P_A^2 = P_A = P_A^*$. □

Three more exact cases mark the edge of the record [T]:

- $C = pI$ has full raw census for $p \in [\varepsilon, 1 - \varepsilon]$, and zero boundary census for every p.
- $\frac{1}{2}(|10\rangle\langle 10| + |01\rangle\langle 01|)$ has $C = \frac{1}{2}I_2$ and $X_A = 0$, yet covariance $-\frac{1}{4}$ and mutual information $\log 2$.
- $(|1100\rangle + |0011\rangle)/\sqrt{2}$ split $\{1, 2\}|\{3, 4\}$ has $X_A = 0$ and entanglement entropy $\log 2$.

A one-body record does not in general determine correlations outside the quasi-free class.

Proposition 3.6 (number covariance)**.** *For a gauge-invariant quasi-free state,* $\mathrm{Var}(N_A) = \operatorname{Tr} F_A$ *and* $\mathrm{Cov}(N_A, N_B) = -\operatorname{Tr} X_A$. *On a pure Slater state,* $\mathrm{Var}(N_A) = \operatorname{Tr} X_A$. *[T]*

Proof. By Wick's theorem, $\mathrm{Cov}(n_i, n_j) = \delta_{ij} C_{ii} - |C_{ij}|^2$. Sum over $A \times A$ and over $A \times B$. □

Theorem 3.7 (the weight between bounds the information across the cut)**.** *For the same states, with $h(x) = -x \log x - (1 - x)\log(1 - x)$ and $I_G = \operatorname{Tr} h(A_C) + \operatorname{Tr} h(D_C) - \operatorname{Tr} h(C)$:*

$$4 \operatorname{Tr} X_A \le I_G(A : B),$$

and the constant is sharp. If $\delta I \le C \le (1 - \delta)I$, then $I_G \le \operatorname{Tr} X_A/(\delta(1 - \delta))$. Without such a margin there is no upper constant. [T]

Proof. Let $S = I_A \oplus (-I_B)$ and $C_0 = \frac{1}{2}(C + SCS) = A_C \oplus D_C$. Trace Jensen states that for concave f, $\operatorname{Tr} f(\frac{1}{2}(H+K)) \geq \frac{1}{2}\operatorname{Tr} f(H) + \frac{1}{2}\operatorname{Tr} f(K)$. To prove it, diagonalise the mean in a basis u_i, then use scalar concavity and $f(\langle u, Hu\rangle) \geq \langle u, f(H)u\rangle$.

Apply it to $f = h + 2x^2$, which has $f'' = 4 - 1/(x(1-x)) \leq 0$. With unitary invariance under S, this gives $I_G \geq 2(\operatorname{Tr} C^2 - \operatorname{Tr} C_0^2) = 4\operatorname{Tr} X_A$. On $[\delta, 1-\delta]$ the function $h + x^2/(2\delta(1-\delta))$ is convex, and the same argument gives the upper bound.

Sharpness: $C = \frac{1}{2}I + r\sigma_x$ has $I_G = 4r^2 + O(r^4)$ and $\operatorname{Tr} X_A = r^2$. For the failure without a margin, pure Slater states with an occupation tending to 0 have $I_G / \operatorname{Tr} X_A$ unbounded. □

Theorem 3.8 (the offset speed limit)**.** *Let $P(t)$ be a smooth family of orthogonal projectors and E an isometry onto the region. Set $C = E^\dagger P E$ with $0 < C < I$, $g = \frac{1}{2}\|\dot{P}\|_F^2$ and $S = \operatorname{Tr}[(C(I-C))^{-1}]$. Then $\dot{\tau}^2 \leq 4gS$, with equality iff $\dot{P} \propto [P,[P,F]]$, where $F = EWE^\dagger$ and $W = -(C(I-C))^{-1}$. For $\dot{P} = -i[H,P]$, $g = \|PH(1-P)\|_F^2$. [T]*

Proof. $\dot{\tau} = \operatorname{Tr}[W\dot{C}] = \operatorname{Tr}[F\dot{P}]$. Differentiating $P^2 = P$ shows that $\dot{P}$ is block off-diagonal, so only $T = [P,[P,F]]$ pairs with it. Cauchy–Schwarz gives $\dot{\tau}^2 \leq \|T\|_F^2 \|\dot{P}\|_F^2$. Finally, $\|T\|_F^2 = 2\operatorname{Tr}[W(I-C)WC] = 2S$, because W and C commute. For the corollary, $[H,P]$ has two adjoint blocks. □

Proposition 3.9 (the speed budget does not choose the future)**.** *Take $H = \omega\sigma_x$ and $P_\pm(0) = \frac{1}{2}(I \pm \sigma_y)$, read through the first basis vector. Both have $\nu = \frac{1}{2}$, $\tau = 0$, $X = \frac{1}{4}$ and $g = \omega^2$. Then $\tau_\pm(t) = \mp 2\operatorname{artanh}(\sin 2\omega t)$: both saturate $\dot{\tau}^2 = 4gS$ wherever $0 < \nu(t) < 1$, that is away from $\sin 2\omega t = \pm 1$, in opposite directions. [T][V]*

Proof. The Bloch vector $(0, \pm 1, 0)$ rotates in the y–z plane at rate 2ω. With $s = \sin 2\omega t$, $S = 4/(1-s^2)$ and $\dot{\tau}^2 = 16\omega^2/(1-s^2) = 4gS$. □

The cross weight of a fixed cut is not the generator coupling of the moving projector. At $t = 0$, $2\sqrt{\operatorname{Tr} X \cdot S} = 2$ while $|\dot{\tau}| = 4\omega$. Script `s16_weight_between_checks_v2` checks the saturation to 5.9×10^{-15}. On 400 random contractions it checks the defect identity, the two order diagnostics, the positive-spectrum agreement, the ordered-eigenvalue comparison and the norm-stability estimate. On 64 interior covariances it checks the two information inequalities. It also checks the sharp family, whose ratio runs 4.00027, 4.0000027, 4.00000002, and 100 orthogonal projectors (pure Slater states), on which $X_A = A_C - A_C^2$ holds to 6.8×10^{-16}. All checked bounds hold [V].

4 Opened matter

Take the PT-symmetric SSH chain, cells $c \in \mathbb{Z}$, two sites a, b per cell:

$$h(k) = \begin{pmatrix} iu & v + we^{-ik} \\ v + we^{ik} & -iu \end{pmatrix}, \qquad E(k)^2 = v^2 + w^2 - u^2 + 2vw\cos k,$$

with $u, v, w > 0$, $v < w$, and the gap condition $E(\pi)^2 = (w-v)^2 - u^2 > 0$. In real space H has $\pm iu$ on the a and b sites, v inside a cell and w from (c, b) to $(c+1, a)$. Then $H^T = H$, but $H^\dagger \neq H$. Fill the lower band with the biorthogonal projector $P(k) = \frac{1}{2}(I - h(k)/E(k))$, $E > 0$, and let C_R be its block on the half-chain $c \geq 0$.

Theorem 4.1 (what survives opening)**.** *For any idempotent* $C = \begin{pmatrix} A & B \\ E & D \end{pmatrix}$, $Y_A = BE = A - A^2$, *and its eigenvalues are* $\nu_j(1-\nu_j)$ *with algebraic multiplicity. So the radial census* $\#\{|1-2\nu_j| \le r\}$ *survives for every* r*. Positivity does not:* $\begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix}$ *is idempotent with* $Y_A = -2$*. [T][L]*

Proof. The AA block of $C^2 = C$ gives the first identity. Triangularising A gives the eigenvalues. □

Proposition 4.2 (the Petermann line)**.** *The band projectors have* $K(k) = \operatorname{Tr} P^\dagger P = 1 + u^2/E(k)^2$*. The band average is*

$$\langle K \rangle = 1 + \frac{u^2}{E(0)E(\pi)},$$

so $\langle K \rangle = 2$ *iff* $E(0)E(\pi) = u^2$*, iff*

$$(w^2 - v^2)^2 = 2u^2(w^2 + v^2). \tag{2}$$

[T][L]

Proof. $\operatorname{Tr} P^\dagger P = \frac{1}{2}(1 + \sum |d_j|^2/|E|^2)$ for $h = d \cdot \sigma$. Here $\sum |d_j|^2 = |v + we^{ik}|^2 + u^2$ and $|E|^2 = |v + we^{ik}|^2 - u^2$. The average is $\frac{1}{2\pi}\int dk/(A + B\cos k) = (A^2 - B^2)^{-1/2}$ with $A^2 - B^2 = E(0)^2E(\pi)^2$. Squaring $E(0)^2E(\pi)^2 = u^4$ gives $((v+w)^2-u^2)((w-v)^2-u^2) = u^4$, that is $(w^2-v^2)^2-2u^2(w^2+v^2) = 0$. □

At $w = 1$, (2) reads $u_{\text{exit}} = (1-v^2)/\sqrt{2(1+v^2)}$. At $u = 0.5$ this gives $v_{\text{exit}} = 0.4682$. Predicted from $\langle K \rangle = 2$ before the run, the exits at $u = 0.25,\ 0.45,\ 0.65$ are $v = 0.7474014,\ 0.5291503,\ 0.2357262$. The measured census exits of a 40-cell block are 0.7474015, 0.5291503, 0.2357262, differences of about 5×10^{-8} or less in these runs [V] (script `s05b`). The measurement uses whole-cell blocks and an endpoint filter that discards levels with ν within 10^{-9} of 0 or 1. Why the census leaves there is the question the next section answers.

5 The exit ladder

5.1 The Hermitian band inside the opened one

Lemma 5.1. *Put* $E_0 = E(0)$*,* $E_\pi = E(\pi)$*,* $w' = \frac{1}{2}(E_0 + E_\pi)$*,* $v' = \frac{1}{2}(E_0 - E_\pi)$*. Then* $v'^2 + w'^2 = v^2 + w^2 - u^2$*,* $v'w' = vw$ *and*

$$E(k)^2 = |v' + w'e^{ik}|^2,$$

the band of a Hermitian dimerized chain with hoppings $v' < w'$*. Its modulus is* $k = v'/w'$*, and* $k' = \sqrt{1-k^2} = \sqrt{E_0E_\pi}/w'$*. [T]*

Proof. $E_0^2 + E_\pi^2 = 2(v^2+w^2-u^2)$ and $E_0^2 - E_\pi^2 = 4vw$. Then $v'^2+w'^2 = \frac{1}{2}(E_0^2+E_\pi^2)$, $v'w' = \frac{1}{4}(E_0^2-E_\pi^2)$ and $w'^2 - v'^2 = E_0E_\pi$. □

5.2 An operator that commutes with the half-chain block

Let X be the cell-position operator, $X_{(c,\alpha),(c',\beta)} = c\,\delta_{cc'}\delta_{\alpha\beta}$.

Theorem 5.2 (the commuting operator)**.** *Let $T = \frac{1}{2}\{X+\frac{1}{2}, H\}$. Then $[P, T] = 0$, and $T = T_L \oplus T_R$ has no entries across the cut between cells -1 and 0. Consequently*

$$[C_R, T_R] = 0,$$

entrywise, as an identity of infinite matrices. T_R is tridiagonal, with

$$T_R\big|_{(c,a),(c,a)} = iu(c+\tfrac{1}{2}), \qquad T_R\big|_{(c,b),(c,b)} = -iu(c+\tfrac{1}{2}),$$
$$T_R\big|_{(c,a),(c,b)} = v(c+\tfrac{1}{2}), \qquad T_R\big|_{(c,b),(c+1,a)} = w(c+1),$$

and it is symmetric. [T]

Proof. The gap condition gives $E(k)^2 \geq E_\pi^2 > 0$, so $P(k)$ is analytic and periodic. The real-space entries $P_{cc'} = \frac{1}{2\pi}\int e^{ik(c-c')}P(k)\,dk$ therefore decay exponentially, and every product below is an absolutely convergent sum. Integration by parts gives $[X, P]_{cc'} = (c-c')P_{cc'} = \frac{i}{2\pi}\int e^{ik(c-c')}\partial_k P(k)\,dk$. Since $[P, H] = 0$,

$$[P, \{X, H\}] = \{[P, X], H\} \iff -i\{\partial_k P, h\} = \tfrac{i}{2}\{\partial_k(h/E), h\}.$$

Differentiating $(h/E)^2 = I$ gives $\{\partial_k(h/E), h/E\} = 0$, so the right side vanishes. The shift by $\frac{1}{2}$ commutes with P trivially, so $[P, T] = 0$.

The entry $T_{ij} = \frac{1}{2}(x_i + x_j + 1)H_{ij}$ crosses the cut only for $i = (-1, b)$, $j = (0, a)$, where $x_i + x_j + 1 = 0$. So $T = T_L \oplus T_R$. Then $(PT)_{RR} = C_R T_R + P_{RL}T_{LR} = C_R T_R$ and $(TP)_{RR} = T_R C_R$. The listed entries follow from those of H. □

The construction is that of Peschel and of Eisler and Peschel for Hermitian chains [1, 2, 4, 5]. What is new here is that it uses only $h(k)^2 = E(k)^2 I$. That relation survives the non-Hermitian opening, so the commuting operator survives with it. On the same PT-symmetric SSH model, Rottoli, Fossati and Calabrese find numerically a linearly weighted lattice entanglement Hamiltonian in a gapped setting [6]; their hopping and cut conventions differ from those used here and must be translated before sectors are compared. Biorthogonal entanglement spectra of non-Hermitian free fermions are developed in [7, 8]. This paper adds the exact entrywise commutation proof, valid without Hermiticity, the elliptic (Abel) phase tested against the edge levels, and its exact crossing on the Petermann line.

To pass from the entrywise identity to eigenvectors, take T_R as its maximal operator on ℓ^2, with domain $\{r \in \ell^2 : T_R r \in \ell^2\}$. For an eigenvector $r \in \ell^2$ of T_R, the kernel of C_R decays exponentially and the entries of T_R grow only linearly, so the sums defining $T_R C_R r$ and $C_R T_R r$ converge absolutely and agree. Hence C_R maps each eigenspace of T_R into itself, and a simple eigenvector of T_R is an eigenvector of C_R. No completeness or diagonalizability of T_R is claimed.

5.3 The ladder

Lemma (the domain of the phase)**.** *For $0 \leq v < w$ and $0 < u < w - v$, with v' as in Lemma 5.1, the number $s = uw/(w^2 - v'^2)$ satisfies $0 < s < 1$, and $s \to 1$ at the critical boundary $u \to w - v$. [T]*

Proof. At $v = 0$, $E_0 = E_\pi$, so $v' = 0$ and $s = u/w < 1$. For fixed $0 < v < w$, write $v'(u) = (E_0 - E_\pi)/2$. Then

$$\frac{dv'}{du} = \frac{u}{2}\Big(\frac{1}{E_\pi} - \frac{1}{E_0}\Big) > 0,$$

since $E_\pi < E_0$. As u increases to $w - v$, $E_\pi \to 0$ and $E_0 \to 2\sqrt{vw}$, so $v' \to \sqrt{vw}$. Therefore $v'^2 < vw$ in the open gapped region, and

$$w^2 - v'^2 > w(w - v) > wu > 0, \qquad 0 < s = \frac{uw}{w^2 - v'^2} < 1.$$

At the critical boundary $w^2 - v'^2 \to w(w - v)$ and $s \to 1$. □

Measured statement 5.3 (the edge ladder)**.** Let $K = K(k)$ and $K' = K(k')$ be the complete elliptic integrals of the first kind for the modulus of Lemma 5.1. Put

$$s = \frac{uw}{w^2 - v'^2} \in (0, 1) \text{ (by the domain lemma)}, \qquad \theta = \frac{\pi}{K} F(\arcsin s \,|\, k) \in (0, \pi), \qquad \Delta = \frac{2\pi K'}{K}.$$

Equivalently, $\mathrm{sn}(\theta K/\pi \,|\, k) = s$. Then the levels of C_R on the eigenvectors of T_R localised at the cut are

$$\varepsilon_j = \log \frac{1 - \nu_j}{\nu_j} = j\Delta + i\theta, \qquad j \in \mathbb{Z}.$$

All levels share one phase, and there is no real offset. [V]

Evidence (script `s15_exit_ladder_v2`). The test uses 200 random points with $w = 1$, $v \in (0.02, 0.7)$, $u \in (0.02, 0.9)$ and gap $E_\pi \geq 0.2$, of which 16 lie past the exit line. The block has 90 cells, inside a 6000-momentum chain. Results:

- the eigenvector residual $\|C_R r - \nu r\|$ is at most 8.4×10^{-12} over 600 selected levels, three per point ($j = 0, \pm 1$); every point contributes all three, and the residual filter removes none. This is a residual on selected eigenvectors, not an operator norm of $[C_R, T_R]$;
- the relative spacing error is at most 1.7×10^{-8} over 400 spacing pairs, none skipped by the script's $\Delta \geq 40$ rule;
- the phase error is at most 4.2×10^{-7} on the levels $j = 0, \pm 1$;
- the domain condition $0 < s < 1$ holds on the whole gapped region by the domain lemma; at the test points the maximum is 0.913.

Deeper levels have ν within 10^{-10} of 0 or 1. There the nonnormal eigenproblem is ill-conditioned, and they are not used. At $u = 0$, $k = v/w$ and the spacing is the elliptic spacing of the Hermitian dimerized chain known from corner-transfer-matrix methods [2, 3], with the protected $\nu = \frac{1}{2}$ edge level at $j = 0$ [R]. Opening the chain keeps the spacing of the hidden Hermitian band and rotates the whole ladder by one phase.

5.4 The exit theorem

Theorem 5.4 (the exit ladder)**.** *(a) (one phase, one crossing) If $\varepsilon_j = j\Delta + i\theta$ with $\theta \in (0, \pi)$, then every $\Gamma_j = \tanh(\varepsilon_j/2)$ lies on the circular arc through -1, $i\tan(\theta/2)$ and 1. Either all $|\Gamma_j| < 1$, or all $|\Gamma_j| = 1$, or all $|\Gamma_j| > 1$, according as $\theta < \pi/2$, $\theta = \pi/2$ or $\theta > \pi/2$.*

(b) (the crossing is the Petermann line) For θ as in Statement 5.3,

$$\theta = \frac{\pi}{2} \iff E(0)E(\pi) = u^2 \iff \langle K \rangle = 2 \iff (w^2 - v^2)^2 = 2u^2(w^2 + v^2).$$

Moreover $\theta < \pi/2$ on the side $E(0)E(\pi) > u^2$ and $\theta > \pi/2$ on the other.

(c) (the dimer limit, exact) At $v = 0$ the half-chain block has exactly one level off $\{0, 1\}$, with $\varepsilon = 2i \arcsin(u/w)$. This equals the formula of Statement 5.3 at $v = 0$. That level leaves the disk exactly at $w^2 = 2u^2$, which is (2) at $v = 0$.

[T]

Proof. (a) By Theorem 2.2, $|\tanh(W/2)| < 1 \iff \cos \operatorname{Im} W > 0$. Here $\operatorname{Im} \varepsilon_j = \theta$ for every j, in the principal range. The image of the line $\{x + i\theta\}$ under $W \mapsto \tanh(W/2)$ is a circle arc. Its endpoints are ± 1 as $x \to \pm\infty$, and $x = 0$ gives $\tanh(i\theta/2) = i \tan(\theta/2)$.

(b) On $(0, K)$, $\operatorname{sn}(\cdot \,|\, k)$ increases strictly from 0 to 1, and $\operatorname{sn}(K/2 \,|\, k)^2 = 1/(1 + k')$ [9]. So $\theta = \pi/2$ iff $s^2(1 + k') = 1$. Put $W = w'$ and $r = \sqrt{E_0 E_\pi}$, so that $k' = r/W$ (Lemma 5.1). Then

$$\mathcal{F} := u^2 w^2 (W + r) - W(w^2 - v'^2)^2 = 0.$$

Lemma 5.1 makes w^2 and v^2 the roots of $x^2 - (2W^2 - r^2 + u^2)x + W^2(W^2 - r^2) = 0$. With $\delta = u^2 - r^2$ and $D = \sqrt{4W^2u^2 + \delta^2}$, the larger root ($w > v$) is $w^2 = W^2 + \frac{1}{2}(\delta + D)$, and $w^2 - v'^2 = \frac{1}{2}(u^2 + r^2 + D)$. So $\mathcal{F} = \mathcal{F}(D)$ is a polynomial in D. Multiplying by its conjugate $\mathcal{F}(-D)$ and reducing with $D^2 = 4W^2u^2 + \delta^2$ gives the identity

$$\mathcal{F}(D)\,\mathcal{F}(-D) = W\, r^2 u^2\, (W + r)^2\, (W - r)\, (u^2 - r^2).$$

Since $0 < r < W$ and $u > 0$, $\mathcal{F}(D) = 0$ forces $u^2 = r^2 = E(0)E(\pi)$.

Conversely, on $u^2 = r^2$ we have $\delta = 0$, $D = 2Wu$, $w^2 = W(W + u)$ and $w^2 - v'^2 = u(W + u)$. Then $s^2 = w^2/(W+u)^2 = W/(W+u) = 1/(1+k')$. The remaining equivalences are Proposition 4.2.

For the sides: θ is continuous in (v, u) on the gapped region and equals $\pi/2$ only on the line. Each side of the line is connected, since for fixed v it is an interval in u with continuous endpoints. As $u \to 0$, $s \to 0$ and $\theta \to 0$; at $v = 0$, $\theta = 2 \arcsin(u/w) > \pi/2$ for $u^2 > w^2/2$.

(c) At $v = 0$, H is a direct sum of dimers on $\{(c, b), (c + 1, a)\}$ with matrix $\begin{pmatrix} -iu & w \\ w & iu \end{pmatrix}$ and $E = \sqrt{w^2 - u^2}$. The half-chain $c \geq 0$ contains whole dimers, whose blocks are idempotent with $\nu \in \{0, 1\}$, and the single site $(0, a)$ of the cut dimer. For that site, $\nu = \frac{1}{2}(1 - iu/E)$, so $z = (E + iu)/(E - iu)$ has $|z| = 1$ and $\varepsilon = 2i \arctan(u/E) = 2i \arcsin(u/w)$. At $v = 0$ we have $v' = 0$, $k = 0$, $K = \pi/2$ and $F(\phi \,|\, 0) = \phi$, so $s = u/w$ and $\theta = 2 \arcsin(u/w)$. The level leaves where $2 \arcsin(u/w) = \pi/2$, that is $w^2 = 2u^2$, and $E(0)E(\pi) = w^2 - u^2 = u^2$ there. □

Sympy confirms that $\mathcal{F}$ vanishes on the line, and that the product reduces to the factored form above. Script `s15_exit_ladder_v2` returns $|\theta - \pi/2| \leq 6.7 \times 10^{-16}$ on the line at six values of v, and the dimer limit to 2.2×10^{-16} [V].

5.5 What the theorem explains, and its scope

1. *Why the exit is filter-independent on this chain.* The levels nearest the endpoints, $\operatorname{Re} \varepsilon_j \to \pm\infty$, cross with the centre level. A filter that discards them does not move the crossing until the cutoff is lowered enough to retain numerically ill-conditioned near-endpoint levels. The filter audit bears this out (ring 2000, block 40, $v = 0.5$, $w = 1$). The massless exit holds at $u = 0.4743416$, with $\langle K \rangle = 2.000000002$, for every filter from 10^{-6} through 10^{-9}. It moves only in the seventh digit at 10^{-10} and in the fourth at 10^{-12}, where the cutoff is lowered enough to retain numerically ill-conditioned near-endpoint levels [V] (script `incoming_exit_filter`).

2. *Why the quarter-turn continuation fits only at the exit.* The continued endpoint offset $\Gamma_{\text{end}} = i\gamma$, $\gamma = u/\sqrt{E_0 E_\pi}$, is the apex $i \tan(\theta/2)$ of the arc only at $v = 0$ and on the exit line.

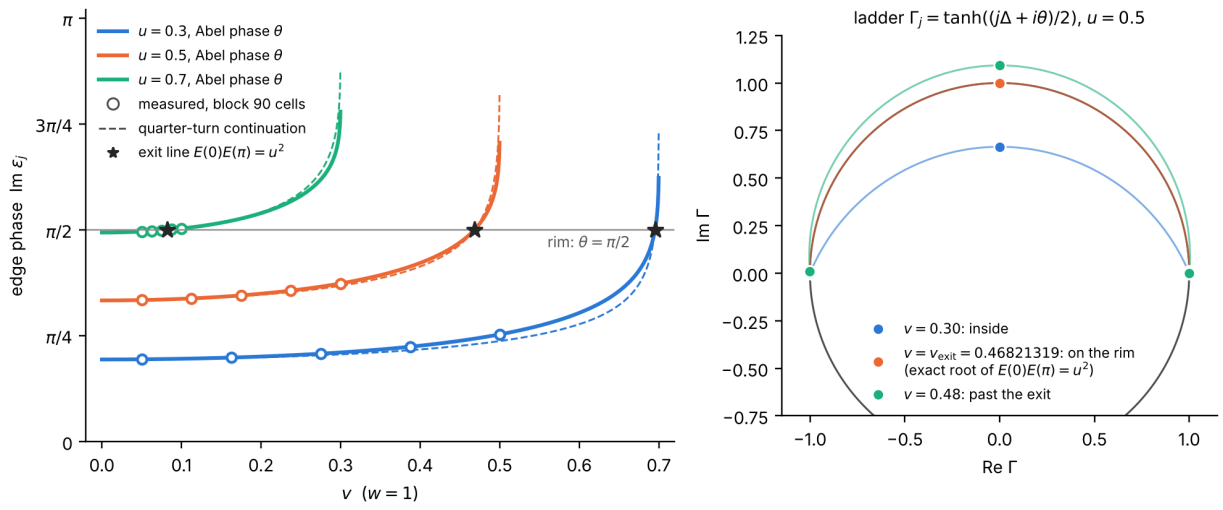


Figure 1: Left: the shared edge phase θ from Statement 5.3 (solid) against v at three values of u ($w = 1$). Open circles: measured $\operatorname{Im}\varepsilon_0$, block of 90 cells. Dashed: the quarter-turn continuation $2\arctan(u/\sqrt{E_0 E_\pi})$. Stars: the exit line (2), where every solid curve crosses $\pi/2$. Right: the ladder Γ_j, $j = -3, \ldots, 3$, at $u = 0.5$, inside, on and past the exit. Each ladder sits on one arc through ± 1, so it crosses the rim all at once.

Elsewhere $\tan(\theta/2) \neq \gamma$ (Figure 1, dashed). At nine points on the approach to the exit ($u = 0.3, 0.5, 0.7$), plain finite blocks measure a census rotation above the continuation by 4×10^{-6} to 3.4×10^{-2}, independent of block size (script `s11_exit_quarter_turn_v2`). That rotation is the Abel phase: at the same nine points and blocks of 40 and 80 cells, the largest census rotation equals θ/π to 7.2×10^{-7} (script `s18`). This is a check on Statement 5.3 that does not use the commuting operator [V].

3. *Partition.* The cut runs between cells. With $w = 0$ every whole-cell block is idempotent and nothing leaves, even where $\langle K \rangle = 2$. That is consistent with (c): the level that exits belongs to a dimer cut in two.

4. *Other bands.* Theorem 5.2 uses only $h(k)^2 = E(k)^2 I$ and nearest-cell hopping, so the commuting operator exists for any such two-band chain. Statement 5.3 and part (b) are stated for real u only, where the on-site term is purely imaginary. The massive band $d = (t_1 + \cos k, \sin k, m + i\gamma)$ has a complex on-site term. At $m = 0.4$ its census exits far earlier, at $\langle K \rangle \approx 1.02$ (script `s06`), and the exit moves with the filter, from $\gamma = 0.2716$ at 10^{-6} to 0.1228 at 10^{-12} (script `incoming_exit_filter`). Whether its ladder still carries a single phase is not tested here.

The open step. Theorem 5.2 supplies invariant eigenspaces of T_R for C_R; completeness or diagonalizability of T_R is not claimed. Theorem 5.4 turns any ladder of the measured form into the exit law. What remains is to derive the ladder form and the elliptic phase of Statement 5.3 from T_R analytically. In the Hermitian case this is done by the corner transfer matrix. Here the natural route is the Wiener–Hopf factorisation of the band symbol on the elliptic curve $y^2 = -z(v' + w'z)(v'z + w')$. On that curve the phase appears as the Abel image of the point $z = -v/w$, where the opened eigenvector $(E - iu, -(v + wz))$ degenerates. The numerical entanglement Hamiltonian of [6] and

the biorthogonal framework of [7, 8] are the natural points of comparison for that derivation. That derivation is open [C]. With it, the exit law on the coupled PT-symmetric chain would be a theorem end to end.

6 Status

Result	Grade
Bridge	[T][L]
Cayley domain (real cases also in Lean)	[T]
Weight between: split, order, two-sidedness, comparison, trace, local unitaries	[T][L] (Lean: algebraic subclaim)
Operator-norm stability, near-projector comparison, Hölder radius	[T]
Projector extension and three exact limits of the one-body record	[T]
$4 \operatorname{Tr} X_A \leq I_G$, sharp; upper bound with margin, none without	[T][V]
Offset speed limit and the opposite-direction saturating pair	[T][V]
Radial census survives opening; positivity does not	[T][L] (Lean: algebraic subclaim)
Petermann line $\langle K \rangle = 2 \iff (w^2 - v^2)^2 = 2u^2(w^2 + v^2)$	[T][L] (Lean: algebraic subclaim)
Hidden Hermitian band; commuting operator $[C_R, T_R] = 0$	[T]
Conditional ladder crossing (any ladder of the stated form) and exact crossing of the proposed phase on the Petermann line	[T]
Dimer limit, exact	[T]
Ladder form $\varepsilon_j = j\Delta + i\theta$, elliptic spacing and Abel phase, 200 random points	[V]
Finite-block census exit on the line, about 5×10^{-8} in these runs; filter audit	[V]
Census rotation of plain blocks equals the Abel phase, to 7.2×10^{-7}	[V]
Analytic derivation of the ladder from T_R	[C]

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A Conventions

Logarithms are natural. $K(k) = F(\pi/2\,|\,k)$ with $F(\phi\,|\,k) = \int_0^\phi (1 - k^2 \sin^2 t)^{-1/2} dt$ (the scripts call `mpmath` with parameter $m = k^2$). The biorthogonal projector uses the principal square root $E(k) > 0$, which is valid throughout the gapped region. The cut lies between cells -1 and 0, and the region is $c \geq 0$; in the scripts it is the left edge of a finite block of 90 cells. The phases $\mathrm{Im}\,\varepsilon$ use the principal logarithm.

B Scripts

Each listing prints the numbers quoted in the text. The outputs are reproduced after each listing.

B.1 `s15_exit_ladder_v2.py`

```
# SCRIPT: s15_exit_ladder_v2
"""s15: the exit ladder on the opened PT-symmetric SSH chain.
Checks, at 200 random gapped parameter points (w=1, seed 15):
  (a) T_R = 1/2{X+1/2,H} restricted to the right half commutes with the biorthogonal block C_R
      (measured as the eigen-residual of C_R on edge-localised eigenvectors of T_R);
  (b) the edge levels are eps_j = j*Delta + i*theta with Delta = 2*pi*K'(k)/K(k), k = v'/w';
  (c) theta = pi*F(arcsin s | k^2)/K(k), s = u*w/(w^2 - v'^2);
  (d) the ladder is inside the disk iff E(0)E(pi) > u^2 (sign of cos theta).
Also the exact dimer case v=0, theta = 2 arcsin(u/w).
v2: asserts that every point contributes exactly 3 selected levels (j=0,+-1) and that the residual
filter removes none; prints the per-point count summary, the number of spacing pairs used and the
number skipped by the Delta>=40 rule."""
import numpy as np, mpmath as mp
rng=np.random.default_rng(15)
def block(v,w,u,Lc,N=6000):
    ks=2*np.pi*(np.arange(N)+0.5)/N
    E=np.sqrt(v*v+w*w-u*u+2*v*w*np.cos(ks)+0j)
    q=v+w*np.exp(-1j*ks) # h = [[iu, q],[conj-in-k q~, -iu]]
    qt=v+w*np.exp(1j*ks)
    P=np.empty((N,2,2),complex) # lower band P = (I - h/E)/2
    P[:,0,0]=0.5*(1-1j*u/E); P[:,1,1]=0.5*(1+1j*u/E)
    P[:,0,1]=-0.5*q/E; P[:,1,0]=-0.5*qt/E
    d=np.arange(-(Lc-1),Lc)
    G=np.einsum('dk,kab->dab',np.exp(1j*np.outer(d,ks)),P)/N
    C=np.zeros((2*Lc,2*Lc),complex)
    for c in range(Lc):
        for cp in range(Lc):
            C[2*c:2*c+2,2*cp:2*cp+2]=G[c-cp+Lc-1]
    return C
def Tmat(v,w,u,L):
    T=np.zeros((2*L,2*L),complex)
    for c in range(L):
        a,b=2*c,2*c+1; x=c+0.5
        T[a,a]=1j*u*x; T[b,b]=-1j*u*x; T[a,b]=T[b,a]=v*x
        if c<L-1: T[b,a+2]=T[a+2,b]=w*(c+1)
    return T
def predict(v,w,u):
    E0=np.sqrt((v+w)**2-u*u); Ep=np.sqrt((w-v)**2-u*u)
    wp=(E0+Ep)/2; vp=(E0-Ep)/2; k=vp/wp
```

```
    K=float(mp.ellipk(k*k)); Kp=float(mp.ellipk(1-k*k))
    s=u*w/(w*w-vp*vp)
    th=np.pi*float(mp.ellipf(mp.asin(s),k*k))/K
    return 2*np.pi*Kp/K, th, E0*Ep
def measure(v,w,u,L=90):
    C=block(v,w,u,L); T=Tmat(v,w,u,L)
    ev,R=np.linalg.eig(T); lv=[]
    for i in np.argsort(abs(ev))[:12]:
        r=R[:,i]/np.linalg.norm(R[:,i])
        if np.abs(r[-30:]).max()>1e-9: continue
        nu=(r.conj()@C@r)
        if abs(nu)<1e-14 or abs(1-nu)<1e-14: continue
        res=np.linalg.norm(C@r-nu*r)
        lv.append((ev[i].real,np.log((1-nu)/nu),res))
    lv.sort(key=lambda x:x[0])
    return lv
out=[]
pts=[]
while len(pts)<200:
    v=rng.uniform(0.02,0.7); u=rng.uniform(0.02,0.9)
    Ep2=(1-v)**2-u*u
    if Ep2<0.2**2 : continue
    pts.append((v,u))
worst={'res':0,'phase':0,'spacing':0}; agree=0; outside=0
counts={}; pairs_used=0; pairs_skipped=0; dropped_by_residual=0
for (v,u) in pts:
    D,th,p=predict(v,1.0,u)
    lv=measure(v,1.0,u)
    assert lv, f'no␣edge␣levels␣at␣v={v},␣u={u}'
    j0=min(range(len(lv)),key=lambda i:abs(lv[i][1].real))
    cand=[x for x in lv if abs(x[1].real)<1.5*D+1e-9] # j=0,+-1
    use=[x for x in cand if x[2]<1e-6]
    dropped_by_residual+=len(cand)-len(use)
    assert len(cand)==len(use), f'residual␣filter␣dropped␣a␣level␣at␣v={v},␣u={u}'
    assert len(use)==3, f'{len(use)}␣selected␣levels␣(expected␣3)␣at␣v={v},␣u={u}'
    counts[len(use)]=counts.get(len(use),0)+1
    for t,e,res in use:
        worst['res']=max(worst['res'],res)
        worst['phase']=max(worst['phase'],abs(np.angle(np.exp(1j*(e.imag-th)))))
    re=sorted(x[1].real for x in use)
    for a,b in zip(re,re[1:]):
        if D<40:
            worst['spacing']=max(worst['spacing'],abs((b-a)-D)/D); pairs_used+=1
        else: pairs_skipped+=1
    meas_in=np.cos(lv[j0][1].imag)>0; pred_in=p>u*u
    agree+= (meas_in==pred_in); outside+= (not pred_in)
lines=[f'points:␣200␣(w=1,␣seed␣15,␣gap␣E(pi)>=0.2,␣block␣90␣cells),␣of␣which␣{outside}␣lie␣past␣the␣exit␣line
    ␣E0*Epi<u^2',
 f'(a)␣max␣eigen-residual␣of␣C_R␣on␣T_R␣edge␣eigenvectors␣(levels␣j=0,+-1):␣{worst["res"]:.2e}',
 f'(b)␣max␣relative␣spacing␣error␣|Re(eps_(j+1)-eps_j)-Delta|/Delta:␣␣␣␣␣␣␣␣␣{worst["spacing"]:.2e}',
 f'(c)␣max␣phase␣error␣|Im␣eps_j␣-␣theta_pred|␣(j=0,+-1):␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣{worst["phase"]:.2e}',
 f'(d)␣inside/outside␣verdict␣agrees␣with␣sign(E0*Epi-u^2):␣{agree}/200',
 'selected␣levels␣per␣point␣(count:␣number␣of␣points):␣'+',␣'.join(f'{k}:␣{n}' for k,n in sorted(counts.items
    ()))+f';␣total␣selected␣levels␣{sum(k*n␣for␣k,n␣in␣counts.items())}',
 f'levels␣removed␣by␣the␣residual␣filter␣(res␣>=␣1e-6):␣{dropped_by_residual}',
 f'spacing␣pairs␣used␣in␣(b):␣{pairs_used};␣skipped␣by␣the␣Delta>=40␣rule:␣{pairs_skipped}']
# exact dimer case v=0
md=0
for u in [0.1,0.3,0.5,0.6,0.7,0.75,0.9]:
    D,th,p=predict(0.0,1.0,u)
    md=max(md,abs(th-2*np.arcsin(u)))
lines.append(f'v=0␣dimer␣limit:␣max␣|theta_pred␣-␣2␣arcsin(u/w)|␣over␣7␣values␣of␣u:␣{md:.2e}')
# exit line: theta_pred at E0*Epi=u^2 equals pi/2
mx=0
for v in [0.05,0.1,0.2,0.3,0.4,0.5]:
    uu=(1-v*v)/np.sqrt(2*(1+v*v)) # closed form of E0*Epi=u^2 at w=1
    mx=max(mx,abs(predict(v,1.0,uu)[1]-np.pi/2))
lines.append(f'exit␣line:␣max␣|theta_pred␣-␣pi/2|␣on␣E0*Epi=u^2␣at␣6␣values␣of␣v:␣{mx:.2e}')
```

```
smax=0
for (v,u) in pts:
    E0=np.sqrt((1+v)**2-u*u); Ep=np.sqrt((1-v)**2-u*u); vp=(E0-Ep)/2
    smax=max(smax,u/(1-vp*vp))
lines.append(f'domain: max s = u*w/(w^2-v\'^2) over the 200 points: {smax:.4f} (< 1)')
print('\n'.join(lines))
open('../outputs/s15_exit_ladder_v2.txt','w').write('\n'.join(lines)+'\n')
```

```
points: 200 (w=1, seed 15, gap E(pi)>=0.2, block 90 cells), of which 16 lie past the exit line E0*Epi<u^2
(a) max eigen-residual of C_R on T_R edge eigenvectors (levels j=0,+-1): 8.37e-12
(b) max relative spacing error |Re(eps_(j+1)-eps_j)-Delta|/Delta: 1.67e-08
(c) max phase error |Im eps_j - theta_pred| (j=0,+-1): 4.21e-07
(d) inside/outside verdict agrees with sign(E0*Epi-u^2): 200/200
selected levels per point (count: number of points): 3: 200; total selected levels 600
levels removed by the residual filter (res >= 1e-6): 0
spacing pairs used in (b): 400; skipped by the Delta>=40 rule: 0
v=0 dimer limit: max |theta_pred - 2 arcsin(u/w)| over 7 values of u: 2.22e-16
exit line: max |theta_pred - pi/2| on E0*Epi=u^2 at 6 values of v: 6.66e-16
domain: max s = u*w/(w^2-v'^2) over the 200 points: 0.9131 (< 1)
```

B.2 s16_weight_between_checks_v2.py

```
# SCRIPT: s16_weight_between_checks_v2
"""s16: numerical controls for the weight-between theorems (journal edition).
Random positive contractions C on a=4, b=5 (seed 16)."""
import numpy as np
rng=np.random.default_rng(16); a,b=4,5; n=a+b
def rand_contraction(n,pure=False):
    Z=rng.normal(size=(n,n))+1j*rng.normal(size=(n,n)); U,_=np.linalg.qr(Z)
    lam=(rng.integers(0,2,n).astype(float) if pure else rng.uniform(0,1,n))
    return U@np.diag(lam)@U.conj().T
h=lambda x: -(x*np.log(x)+(1-x)*np.log(1-x))
def trh(M): e=np.linalg.eigvalsh(M); e=np.clip(e,1e-300,1-1e-16); return np.sum(h(e))
L=[]; w={}
def upd(k,x): w[k]=max(w.get(k,-np.inf),x)
for _ in range(400):
    C=rand_contraction(n); A=C[:a,:a]; B=C[:a,a:]; D=C[a:,a:]
    X=B@B.conj().T; XB=B.conj().T@B; F=A-A@A
    TA=(C-C@C)[:a,:a]
    upd('defect split |F-T-X|',np.abs(F-TA-X).max())
    ex=np.linalg.eigvalsh(X); ef=np.linalg.eigvalsh(F)
    upd('order: max eig X - 1/4',ex.max()-0.25); upd('order: min eig(F-X)',-np.linalg.eigvalsh(F-X).min())
    pos=lambda e: np.sort(e[e>1e-12])
    upd('two-sided |spec X_A - spec X_B|',np.abs(pos(ex)-pos(np.linalg.eigvalsh(XB))).max())
    # min-max comparison
    upd('ordered lambda_j(X)-lambda_j(F)',(np.sort(ex)-np.sort(ef)).max())
    # stability
    E=rng.normal(size=(n,n))+1j*rng.normal(size=(n,n)); E=(E+E.conj().T)/2; E*=1e-3/np.linalg.norm(E,2)
    Ct=C+E; ev,V=np.linalg.eigh(Ct); Ct=V@np.diag(np.clip(ev,0,1))@V.conj().T
    dl=np.linalg.norm(C-Ct,2); Xt=Ct[:a,a:]@Ct[:a,a:].conj().T
    upd('stability ||X-X~||/delta - 1',np.linalg.norm(X-Xt,2)/dl-1)
# information bound on 64 interior covariances
lo=np.inf; up=np.inf
for _ in range(64):
    C=rand_contraction(n); ev,V=np.linalg.eigh(C); dlt=0.05; ev=dlt+(1-2*dlt)*ev; C=V@np.diag(ev)@V.conj().T
    A=C[:a,:a]; D=C[a:,a:]; X=C[:a,a:]@C[:a,a:].conj().T
    I=trh(A)+trh(D)-trh(C); tx=np.trace(X).real
    lo=min(lo,I-4*tx); up=min(up,tx/(dlt*(1-dlt))-I)
L.append(f'info bound, 64 covariances with spectrum in [0.05,0.95]: min(I_G-4TrX)={lo:.3e}, min(TrX/(d(1-d))-
    I_G)={up:.3e}')
for r in [1e-2,1e-3,1e-4]:
    I=2*np.log(2)-2*h(0.5+r); L.append(f'sharp family C=1/2 I+r sigma_x, r={r:g}: I_G/TrX={I/r**2:.9f}')
# orthogonal projectors (pure Slater states): census recovery
mx=0
for _ in range(100):
```

```
    C=rand_contraction(n,pure=True); A=C[:a,:a]; X=C[:a,a:]@C[:a,a:].conj().T
    mx=max(mx,np.abs(X-(A-A@A)).max())
L.append(f'orthogonal projectors (pure Slater states) (100): max |X_A-(A_C-A_C^2)| = {mx:.2e}')
# speed-limit saturating pair
sx=np.array([[0,1],[1,0]],complex); sy=np.array([[0,-1j],[1j,0]])
worst=0
for om in [0.5,1.0,2.0]:
    H=om*sx
    for sgn in [+1,-1]:
        P0=0.5*(np.eye(2)+sgn*sy)
        for t in np.linspace(0.01,0.7/om,31):
            if abs(np.sin(2*om*t))>0.95: continue
            U=np.cos(om*t)*np.eye(2)-1j*np.sin(om*t)*sx; P=U@P0@U.conj().T
            Pd=-1j*(H@P-P@H); nu=P[0,0].real; nud=Pd[0,0].real
            taud=-nud/(nu*(1-nu)); g=0.5*np.linalg.norm(Pd)**2; S=1/(nu*(1-nu))
            worst=max(worst,abs(taud**2/(4*g*S)-1))
L.append(f'offset speed limit, pair P+-: max |taudot^2/(4gS)-1| over 3 omegas, 2 signs, 31 times = {worst:.1e}
    ')
for k,x in w.items(): L.insert(0,f'400 random contractions (a=4,b=5): max[{k}] = {x:.2e}')
L.insert(0,'(for order and stability lines, a negative maximum means the bound holds with that margin)')
print('\n'.join(L)); open('../outputs/s16_weight_between_checks_v2.txt','w').write('\n'.join(L)+'\n')
```

```
(for order and stability lines, a negative maximum means the bound holds with that margin)
400 random contractions (a=4,b=5): max[stability ||X-X~||/delta - 1] = -5.66e-01
400 random contractions (a=4,b=5): max[ordered lambda_j(X)-lambda_j(F)] = -2.54e-02
400 random contractions (a=4,b=5): max[two-sided |spec X_A - spec X_B|] = 1.39e-16
400 random contractions (a=4,b=5): max[order: min eig(F-X)] = -2.26e-02
400 random contractions (a=4,b=5): max[order: max eig X - 1/4] = -5.72e-02
400 random contractions (a=4,b=5): max[defect split |F-T-X|] = 1.87e-16
info bound, 64 covariances with spectrum in [0.05,0.95]: min(I_G-4TrX)=2.106e-03, min(TrX/(d(1-d))-I_G)=3.318e
    -01
sharp family C=1/2 I+r sigma_x, r=0.01: I_G/TrX=4.000266709
sharp family C=1/2 I+r sigma_x, r=0.001: I_G/TrX=4.000002667
sharp family C=1/2 I+r sigma_x, r=0.0001: I_G/TrX=4.000000020
orthogonal projectors (pure Slater states) (100): max |X_A-(A_C-A_C^2)| = 6.80e-16
offset speed limit, pair P+-: max |taudot^2/(4gS)-1| over 3 omegas, 2 signs, 31 times = 5.9e-15
```

B.3 fig09_exit_ladder_v2.py

```
# SCRIPT: fig09_exit_ladder_v2
"""fig09: the exit ladder. Left: theta(v) from the Abel formula, measured edge phases, quarter-turn
    continuation.
Right: the ladder Gamma_j = tanh((j*Delta + i*theta)/2) in the disk, inside / on / past the exit line."""
import numpy as np, mpmath as mp, matplotlib
matplotlib.use('Agg'); import matplotlib.pyplot as plt
exec(open('s15_exit_ladder_v2.py').read().split('out=[]')[0])
plt.rcParams.update({'font.size':9,'axes.spines.top':False,'axes.spines.right':False,'pdf.fonttype':42})
cols=['#2a78d6','#eb6834','#1baf7a']
fig,ax=plt.subplots(1,2,figsize=(9.2,3.9),gridspec_kw={'width_ratios':[1.35,1]})
a=ax[0]
for u,c in zip([0.3,0.5,0.7],cols):
    vmax=1-u-1e-4; vs=np.concatenate([np.linspace(0.0,vmax-0.05,150),vmax-np.geomspace(0.05,1e-4,150)])
    th=[predict(v,1,u)[1] for v in vs]
    gam=[2*np.arctan(u/np.sqrt(np.sqrt((1+v)**2-u*u)*np.sqrt((1-v)**2-u*u))) for v in vs]
    a.plot(vs,th,color=c,lw=2,label=f'$u={u}$, Abel phase $\\theta$')
    a.plot(vs,gam,color=c,lw=1,ls='--')
    vm=np.linspace(0.05,1-u-0.2,5)
    for v in vm:
        lv=measure(v,1,u); e=min(lv,key=lambda x:abs(x[1].real))[1]
        a.plot(v,e.imag,'o',ms=5,mfc='white',mec=c,mew=1.5)
    ve=float(mp.findroot(lambda x:(1-x*x)/mp.sqrt(2*(1+x*x))-u,0.4))
    a.plot(ve,np.pi/2,'*',color='0.15',ms=9,zorder=5)
a.axhline(np.pi/2,color='0.55',lw=0.8); a.text(0.56,np.pi/2-0.13,'rim: $\\theta=\\pi/2$',color='0.35',fontsize
    =8)
a.set_xlabel('$v$ ($w=1$)'); a.set_ylabel('edge phase  Im $\\varepsilon_j$')
```

```
a.set_yticks([0,np.pi/4,np.pi/2,3*np.pi/4,np.pi]); a.set_yticklabels(['0','$\\pi/4$','$\\pi/2$','$3\\pi/4$','$
    \\pi$'])
a.plot([],[],'o',mfc='white',mec='0.3',label='measured␣block␣90␣cells'); a.plot([],[],'--',color='0.3',lw=1,
    label='quarter-turn␣continuation')
a.plot([],[],'*',color='0.15',label='exit␣line␣$E(0)E(\\pi)=u^2$')
a.legend(frameon=False,fontsize=7.5,loc='upper␣left'); a.set_ylim(0,np.pi*1.02); a.set_xlim(-0.02,0.72)
b=ax[1]; t=np.linspace(0,2*np.pi,400); b.plot(np.cos(t),np.sin(t),color='0.3',lw=1)
u=0.5
v_exit=float(mp.findroot(lambda x:(1-x*x)/mp.sqrt(2*(1+x*x))-u,0.47)) # exact root of (1-v^2)/sqrt(2(1+v^2)) =
    u
for v,c,lab in [(0.3,cols[0],'$v=0.30$:␣inside'),(v_exit,cols[1],f'$v=v_{{\\rm␣exit}}={v_exit:.8f}$:␣on␣the␣
    rim\n(exact␣root␣of␣$E(0)E(\\pi)=u^2$)'),(0.48,cols[2],'$v=0.48$:␣past␣the␣exit')]:
   D,th,p=predict(v,1,u); x=np.linspace(-12,12,400)
   G=np.tanh((x+1j*th)/2); b.plot(G.real,G.imag,color=c,lw=1,alpha=0.6)
   j=np.arange(-3,4); Gj=np.tanh((j*D+1j*th)/2)
   b.plot(Gj.real,Gj.imag,'o',color=c,ms=6,mec='white',mew=1,label=lab)
b.set_aspect('equal'); b.set_xlim(-1.15,1.15); b.set_ylim(-0.75,1.25)
b.set_xlabel('Re␣$\\Gamma$'); b.set_ylabel('Im␣$\\Gamma$'); b.legend(frameon=False,fontsize=7.5,loc='lower␣
    center',ncol=1)
b.set_title('ladder␣$\\Gamma_j=\\tanh((j\\Delta+i\\theta)/2)$,␣$u=0.5$',fontsize=9)
plt.tight_layout(); plt.savefig('../figures/fig09_exit_ladder_v2.png',dpi=220); plt.savefig('../figures/
    fig09_exit_ladder_v2.pdf')
```

B.4 Symbolic check of Theorem 5.4(b)

```
# SCRIPT: s17_exit_converse_sympy
"""s17: symbolic check of Theorem 5.4(b). W=w', r=sqrt(E0*Epi), D=sqrt(4W^2u^2+(u^2-r^2)^2).
F = u^2 w^2 (W+r) - W (w^2-v'^2)^2 vanishes iff theta = pi/2."""
import sympy as sp
W,r,u,D=sp.symbols('W␣r␣u␣D',positive=True)
d=u**2-r**2
w2=W**2+(d+D)/2; vp2=W**2-r**2
F=sp.expand(u**2*w2*(W+r)-W*(w2-vp2)**2)
prod=sp.expand(sp.expand(F*F.subs(D,-D)).subs(D**2,4*W**2*u**2+d**2))
fac=sp.factor(prod)
target=W*r**2*u**2*(W+r)**2*(W-r)*(u**2-r**2)
lines=[f'F(D)F(-D)␣factors␣as:␣{fac}',
       f'F(D)F(-D)␣-␣W␣r^2␣u^2␣(W+r)^2␣(W-r)(u^2-r^2)␣=␣{sp.simplify(prod-sp.expand(target))}',
       f'F␣on␣the␣line␣u=r␣(D=2Wu):␣{sp.simplify(F.subs(D,2*W*u).subs(r,u))}']
# root-check: w^2 is the larger root of x^2-(2W^2-r^2+u^2)x+W^2(W^2-r^2)
x=sp.symbols('x'); q=x**2-(2*W**2-r**2+u**2)*x+W**2*(W**2-r**2)
lines.append(f'w^2␣satisfies␣the␣quadratic:␣{sp.simplify(sp.expand(q.subs(x,w2)).subs(D**2,4*W**2*u**2+d**2))}
    ')
print('\n'.join(lines)); open('../outputs/s17_exit_converse_sympy.txt','w').write('\n'.join(lines)+'\n')
```

```
F(D)F(-D) factors as: W*r**2*u**2*(-W + r)*(W + r)**2*(r - u)*(r + u)
F(D)F(-D) - W r^2 u^2 (W+r)^2 (W-r)(u^2-r^2) = 0
F on the line u=r (D=2Wu): 0
w^2 satisfies the quadratic: 0
```

B.5 s18_census_rotation_vs_abel.py

```
# SCRIPT: s18_census_rotation_vs_abel
"""s18: independent check of the Abel phase. Plain finite blocks (no commuting operator):
the largest census rotation max|Im eps|/pi over the block's levels, at the nine (u,v) points of s11,
against theta/pi from the elliptic formula. Endpoint filter |nu|,|1-nu| > 1e-9, principal log."""
import numpy as np
exec(open('s15_exit_ladder.py').read().split('out=[]')[0])
pts=[(0.3,0.4171),(0.3,0.5562),(0.3,0.6604),(0.5,0.2809),(0.5,0.3746),(0.5,0.4448),(0.7,0.0492),(0.7,0.0656)
    ,(0.7,0.0779)]
lines=['␣u␣␣␣␣␣v␣␣␣␣␣␣␣␣block␣␣␣census␣max|Im␣eps|/pi␣␣␣␣Abel␣theta/pi␣␣␣␣difference']
worst=0
```

```
for u,v in pts:
    th=predict(v,1.0,u)[1]/np.pi
    for L in (40,80):
        nu=np.linalg.eigvals(block(v,1.0,u,L))
        nu=nu[(np.abs(nu)>1e-9)&(np.abs(1-nu)>1e-9)]
        rot=np.max(np.abs(np.log((1-nu)/nu).imag))/np.pi
        worst=max(worst,abs(rot-th))
        lines.append(f'{u:.1f}␣␣{v:.4f}␣␣␣{L:4d}␣␣␣␣␣␣{rot:.9f}␣␣␣␣␣␣␣␣␣␣{th:.9f}␣␣␣␣␣{rot-th:+.2e}')
lines.append(f'largest␣|difference|␣over␣9␣points␣and␣2␣block␣sizes:␣{worst:.2e}')
print('\n'.join(lines)); open('../outputs/s18_census_rotation_vs_abel.txt','w').write('\n'.join(lines)+'\n')
```

```
 u v block census max|Im eps|/pi Abel theta/pi difference
0.3 0.4171 40 0.230133206 0.230133206 +4.02e-12
0.3 0.4171 80 0.230133216 0.230133206 +1.01e-08
0.3 0.5562 40 0.276925773 0.276925154 +6.19e-07
0.3 0.5562 80 0.276925356 0.276925154 +2.02e-07
0.3 0.6604 40 0.372154619 0.372154620 -1.34e-09
0.3 0.6604 80 0.372154622 0.372154620 +2.54e-09
0.5 0.2809 40 0.367545806 0.367545731 +7.43e-08
0.5 0.2809 80 0.367545731 0.367545731 +8.71e-12
0.5 0.3746 40 0.405669425 0.405669425 +1.12e-12
0.5 0.3746 80 0.405669425 0.405669425 +8.97e-13
0.5 0.4448 40 0.463575343 0.463574889 +4.54e-07
0.5 0.4448 80 0.463575609 0.463574889 +7.21e-07
0.7 0.0492 40 0.495879103 0.495879103 +4.44e-16
0.7 0.0492 80 0.495879103 0.495879103 -1.11e-16
0.7 0.0656 40 0.497660473 0.497660472 +1.32e-09
0.7 0.0656 80 0.497660474 0.497660472 +1.92e-09
0.7 0.0779 40 0.499359220 0.499359218 +1.96e-09
0.7 0.0779 80 0.499359218 0.499359218 +3.89e-16
largest |difference| over 9 points and 2 block sizes: 7.21e-07
```

B.6 s05b_predict_exit.py (with its helper s05a_exit_line.py)

```
# SCRIPT: MATTER-PEEL-03c locate the disk-exit line v_exit(u) and the rotation at the critical point
import numpy as np
exec(open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]); w=1.0
def census(v,u,N=2000,Lc=40,d=1e-9):
    nu=np.linalg.eigvals(corr(v,w,u,N,Lc)).astype(complex)
    keep=(np.abs(nu)>d)&(np.abs(1-nu)>d); nu=nu[keep]
    W=np.log((1-nu)/nu); return np.max(np.abs(W.imag))/np.pi
def vexit(u):
    lo,hi=0.0,1-u # rotation <1/2 at v=0 (check), critical at v=1-u
    if census(lo,u)>=0.5: return None
    for _ in range(30):
        m=(lo+hi)/2
        if census(m,u)<0.5: lo=m
        else: hi=m
    return (lo+hi)/2
print('␣u␣␣␣␣␣␣v_exit␣␣␣␣␣crit=1-u␣␣␣gap␣at␣exit␣|v-w|-u␣␣␣ratio␣u/(w-v_exit)␣␣␣rot␣at␣critical')
for u in [0.2,0.3,0.4,0.5,0.6,0.7]:
    ve=vexit(u); rc=census(1-u,u)
    if ve is None: print(f'␣{u:.2f}␣␣rotation␣already␣>=␣1/2␣at␣v=0;␣rot␣at␣critical␣{rc:.4f}'); continue
    print(f'␣{u:.2f}␣␣{ve:.6f}␣␣␣{1-u:.3f}␣␣␣␣␣{abs(ve-w)-u:.6f}␣␣␣␣␣␣␣␣␣{u/(w-ve):.6f}␣␣␣␣␣␣␣␣␣{rc:.5f}␣pi'
        )
```

```
# SCRIPT: MATTER-PEEL-03d predict the disk-exit line from mean Petermann = 2, then measure
import numpy as np
exec(open('s05a_exit_line.py').read().split("print('␣u␣")[0])
def P(v,u,k):
    H=np.array([[1j*u, v+w*np.exp(-1j*k)],[v+w*np.exp(1j*k), -1j*u]])
    ev,R=np.linalg.eig(H); o=np.argsort(ev.real); R=R[:,o]; L=np.linalg.inv(R).conj().T
    return np.outer(R[:,0],L[:,0].conj())
ks=2*np.pi*(np.arange(4000)+0.5)/4000
```

```
def meanK(v,u): return np.mean([np.linalg.norm(P(v,u,k),2)**2 for k in ks])
def vpred(u):
    lo,hi=0.0,1-u-1e-9
    for _ in range(40):
        m=(lo+hi)/2
        if meanK(m,u)<2: lo=m
        else: hi=m
    return (lo+hi)/2
print('␣u␣␣␣␣␣predicted␣(mean␣K␣=␣2)␣␣␣␣measured␣census␣exit␣␣␣␣difference')
for u in [0.25,0.45,0.65]:
    vp=vpred(u); vm=vexit(u)
    print(f'␣{u:.2f}␣␣␣␣{vp:.7f}␣␣␣␣␣␣␣␣␣␣␣␣{vm:.7f}␣␣␣␣␣␣␣␣␣␣{vm-vp:+.2e}')
```

```
 u predicted (mean K = 2) measured census exit difference
 0.25 0.7474014 0.7474015 +4.04e-08
 0.45 0.5291503 0.5291503 -1.53e-09
 0.65 0.2357262 0.2357262 -3.70e-08
```

B.7 s06_second_model_exit_law.py

```
# SCRIPT: MATTER-DISK-S06 is the disk-exit law universal? second model: the cone paper's band in 1D
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# Section 5 found, on the PT-symmetric SSH chain, that the census levels leave the Smith disk exactly
# where the band-averaged Petermann factor <K> equals 2. For any two-band h(k) = d(k).sigma the
# Petermann factor of the lower band is K(k) = (1 + |d|^2/|E|^2)/2 with E^2 = d.d (checked below).
# A law that belongs to the SSH chain alone would fail on a different band. Here the band is the cone
# paper's family made one-dimensional, d = (t1 + cos k, sin k, m + i*gamma): a mass term, no PT symmetry.
# Path: start closed (gamma = 0, census real, inside the disk) and raise gamma until the census leaves
# the disk (largest rotation |Im ln z| reaches pi/2, census window 1e-9 < nu < 1-1e-9).
# KILL (written before the run): at any of the four (t1, m) settings, <K> at the measured exit differs
# from 2 by more than 1e-4. Control: gamma = 0 must give a real census inside the disk.
import numpy as np
sx=np.array([[0,1],[1,0]],complex); sy=np.array([[0,-1j],[1j,0]]); sz=np.array([[1,0],[0,-1]],complex)
def H(t1,m,g,k): return (t1+np.cos(k))*sx+np.sin(k)*sy+(m+1j*g)*sz
def lower(t1,m,g,k):
    ev,R=np.linalg.eig(H(t1,m,g,k)); o=np.argsort(ev.real); ev,R=ev[o],R[:,o]
    L=np.linalg.inv(R).conj().T; return ev,R,L
def corr(t1,m,g,N,Lc):
    ks=2*np.pi*(np.arange(N)+0.5)/N; C=np.zeros((2*Lc,2*Lc),complex)
    for k in ks:
        ev,R,L=lower(t1,m,g,k); P=np.outer(R[:,0],L[:,0].conj())
        ph=np.exp(1j*k*np.arange(Lc)); C+=np.kron(np.outer(ph,ph.conj()),P)
    return C/N
def rotation(t1,m,g,N=2000,Lc=40,d=1e-9):
    nu=np.linalg.eigvals(corr(t1,m,g,N,Lc)).astype(complex)
    keep=(np.abs(nu)>d)&(np.abs(1-nu)>d); nu=nu[keep]
    if len(nu)==0: return 0.0, 0.0
    W=np.log((1-nu)/nu); return np.max(np.abs(W.imag))/np.pi, np.max(np.abs(1-2*nu))
ks=2*np.pi*(np.arange(4000)+0.5)/4000
def meanK(t1,m,g):
    out=[]
    for k in ks:
        ev,R,L=lower(t1,m,g,k); out.append(np.vdot(L[:,0],L[:,0]).real*np.vdot(R[:,0],R[:,0]).real)
    return np.mean(out)
# check the general Petermann formula on random points
rng=np.random.default_rng(5); worst=0
for _ in range(300):
    t1,m,g,k=rng.uniform(0.2,2),rng.uniform(-1,1),rng.uniform(0,1.5),rng.uniform(0,2*np.pi)
    ev,R,L=lower(t1,m,g,k); d=np.array([t1+np.cos(k),np.sin(k),m+1j*g]); E2=np.sum(d*d)
    if abs(E2)<1e-3: continue
    Kn=np.vdot(L[:,0],L[:,0]).real*np.vdot(R[:,0],R[:,0]).real; Kf=(1+np.sum(abs(d)**2)/abs(E2))/2
    worst=max(worst,abs(Kn-Kf))
print('general␣Petermann␣formula␣K=(1+|d|^2/|E|^2)/2,␣worst␣error␣on␣300␣random␣points:',worst)
print('settings␣(t1,␣m):␣control␣at␣gamma=0,␣measured␣exit␣gamma,␣<K>␣at␣exit')
ok=True
```

```
for t1,m in [(0.5,0.4),(1.5,0.4),(0.5,0.0),(1.5,0.8)]:
    r0,G0=rotation(t1,m,0.0)
    lo,hi=0.0,None; g=0.05
    while hi is None and g<4:
        if rotation(t1,m,g)[0]>=0.5: hi=g
        else: lo=g; g*=1.4
    if hi is None: print(f'  ({t1},{m}): census never left the disk up to gamma=4'); ok=False; continue
    for _ in range(28):
        mid=(lo+hi)/2
        if rotation(t1,m,mid)[0]<0.5: lo=mid
        else: hi=mid
    ge=(lo+hi)/2; K=meanK(t1,m,ge)
    ok&=abs(K-2)<1e-4
    print(f'  ({t1},{m}): control rot={r0:.3f} pi max|Gamma|={G0:.4f} | exit gamma={ge:.7f} |  <K>={K:.6f}  {"
        PASS" if abs(K-2)<1e-4 else "FAIL"}')
print('VERDICT:', 'exit law <K> = 2 holds on the second model' if ok else 'KILLED on the second model at the
    stated bar')
```

```
general Petermann formula K=(1+|d|^2/|E|^2)/2, worst error on 300 random points: 1.4921397450962104e-13
settings (t1, m): control at gamma=0, measured exit gamma, <K> at exit
  (0.5,0.4): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.1691699 | <K>=1.022352 FAIL
  (1.5,0.4): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.1956728 | <K>=1.018838 FAIL
  (0.5,0.0): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.4743416 | <K>=2.000000 PASS
  (1.5,0.8): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.2518871 | <K>=1.015041 FAIL
VERDICT: KILLED on the second model at the stated bar
```

B.8 s11_exit_quarter_turn_v2.py

Its helper nhssh_lib_and_s03a_occupations.py comes first.

```
# SCRIPT: MATTER-PEEL-02B opened matter: NH SSH h(k)=[[iu, v+w e^{-ik}],[v+w e^{ik}, -iu]]
import numpy as np
def bands(v,w,u,k):
    H=np.array([[1j*u, v+w*np.exp(-1j*k)],[v+w*np.exp(1j*k), -1j*u]])
    ev,R=np.linalg.eig(H); o=np.argsort(ev.real); ev,R=ev[o],R[:,o]
    L=np.linalg.inv(R).conj().T # columns: left vectors with <L_m|R_n>=delta
    return ev,R,L
def corr(v,w,u,N,Lc):
    ks=2*np.pi*(np.arange(N)+0.5)/N
    C=np.zeros((2*Lc,2*Lc),complex)
    for k in ks:
        ev,R,L=bands(v,w,u,k)
        P=np.outer(R[:,0],L[:,0].conj()) # occupied lower band, biorthogonal projector
        ph=np.exp(1j*k*np.arange(Lc))
        C+=np.kron(np.outer(ph,ph.conj()),P)
    return C/N
def S_of(nu):
    nu=nu.astype(complex); return -np.sum(nu*np.log(nu)+(1-nu)*np.log(1-nu))
w=1.0
print('B1/B2: biorthogonal occupations -> Gamma=1-2nu, entropy slope')
for (v,u,tag) in [(0.5,0.0,'Hermitian gapped control'),(1.0,0.0,'Hermitian critical control (c=1)'),(0.5,0.3,'
    NH gapped (|v-w|>u)'),(0.5,0.5,'NH critical |v-w|=u (EP)')]:
    N=2000; Ls=[20,40,80,160]; Ss=[]; worst=0; maxim=0
    for Lc in Ls:
        nu=np.linalg.eigvals(corr(v,w,u,N,Lc)); G=1-2*nu
        worst=max(worst,np.max(np.abs(G))); maxim=max(maxim,np.max(np.abs(G.imag)))
        Ss.append(S_of(nu))
    x=np.log(N/np.pi*np.sin(np.pi*np.array(Ls)/N))
    sl=np.polyfit(x,np.real(Ss),1)[0]
    print(f'  {tag:36s} max|Gamma|={worst:9.3e} max|Im Gamma|={maxim:8.2e}  c=3*slope={3*sl:+.3f}  S(Im)~{np.
        max(np.abs(np.imag(Ss))):.1e}')
```

```
# SCRIPT: MATTER-DISK-S11-V2 is the disk exit the quarter-turn of the author's offset?
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# The Offset Belongs to the Boundary gives the endpoint offset of a gapped chain in closed form,
```

```
# 2 artanh(v/sqrt(E_min E_max)). Continued to the opened SSH chain it predicts a complex level offset
# whose imaginary part is Im tau_inf = 2 arctan(u/sqrt(E(0)E(pi))). Then <K> = 1 + u^2/(E(0)E(pi))
# = sec^2(Im tau_inf/2) exactly, so <K> = 2 is the same statement as Im tau_inf = pi/2: a quarter turn.
# If the census rotation (max over block levels of |Im Log((1-nu)/nu)|/pi) is this offset, then the gap
# between measured rotation and 2 arctan(u/sqrt(E0 Epi))/pi must shrink as the block grows.
# KILL (written before the run): at u = 0.3, 0.5, 0.7 and v at 60%, 80%, 95% of the way to the exit,
# the gap at block 160 exceeds the gap at block 20 by more than 1e-9, or exceeds 2e-3 at block 160.
# (v2: non-increase is judged within an absolute slack of 1e-9, so roundoff on a noise-floor gap
# cannot flip PASS/FAIL; strict decrease is not required.)
# OPPOSITE: the same comparison against the plain Hermitian-form guess 2 arctan(u/E(pi))/pi must NOT pass.
import numpy as np
src=open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]; ns={}; exec(src,ns)
w=1.0
def rot(v,u,Lc,N=4000,d=1e-9):
    nu=np.linalg.eigvals(ns['corr'](v,w,u,N,Lc)); nu=nu[(abs(nu)>d)&(abs(1-nu)>d)]
    return np.max(np.abs(np.log((1-nu)/nu).imag))/np.pi
def vexit(u): # E(0)E(pi)=u^2 with E0^2=(v+w)^2-u^2, Epi^2=(w-v)^2-u^2, v<w-u
    from scipy.optimize import brentq
    return brentq(lambda v: np.sqrt(((v+w)**2-u**2)*((w-v)**2-u**2))-u**2, 0.0, w-u-1e-12)
ok=True; okopp=True
for u in [0.3,0.5,0.7]:
    ve=vexit(u)
    for f in [0.6,0.8,0.95]:
        v=f*ve; E0=np.sqrt((v+w)**2-u**2); Ep=np.sqrt((w-v)**2-u**2)
        pred=2*np.arctan(u/np.sqrt(E0*Ep))/np.pi; opp=2*np.arctan(u/Ep)/np.pi
        gaps=[]
        for Lc in [20,40,80,160]:
            gaps.append(rot(v,u,Lc)-pred)
        g=np.abs(gaps); good=(g[-1]<=g[0]+1e-9) and g[-1]<2e-3; ok&=good
        okopp&= not (abs(gaps[-1]+pred-opp)<2e-3)
        print(f'u={u} v={v:.4f} pred={pred:.5f} opposite={opp:.5f} gaps by block 20,40,80,160: '+' '.join(f'{x
            :+.2e}' for x in gaps)+('  PASS' if good else '  FAIL'))
print('VERDICT:','census rotation converges to the offset quarter-turn form' if ok else 'KILLED at the stated
    bar')
print('OPPOSITE (Hermitian-form guess must fail):','fails as required' if okopp else 'ALSO PASSES - test does
    not discriminate')
```

```
u=0.3 v=0.4171 pred=0.22032 opposite=0.34418 gaps by block 20,40,80,160: +9.81e-03 +9.81e-03 +9.81e-03 +9.81e
    -03 FAIL
u=0.3 v=0.5562 pred=0.25556 opposite=0.47249 gaps by block 20,40,80,160: +2.13e-02 +2.13e-02 +2.13e-02 +2.13e
    -02 FAIL
u=0.3 v=0.6604 pred=0.33867 opposite=0.68960 gaps by block 20,40,80,160: +3.35e-02 +3.35e-02 +3.35e-02 +3.35e
    -02 FAIL
u=0.5 v=0.2809 pred=0.36265 opposite=0.48949 gaps by block 20,40,80,160: +4.90e-03 +4.90e-03 +4.90e-03 +4.90e
    -03 FAIL
u=0.5 v=0.3746 pred=0.39764 opposite=0.58975 gaps by block 20,40,80,160: +8.02e-03 +8.02e-03 +8.02e-03 +8.02e
    -03 FAIL
u=0.5 v=0.4448 pred=0.45733 opposite=0.71372 gaps by block 20,40,80,160: +6.25e-03 +6.25e-03 +6.25e-03 +6.25e
    -03 FAIL
u=0.7 v=0.0492 pred=0.49587 opposite=0.52679 gaps by block 20,40,80,160: +9.73e-06 +9.73e-06 +9.73e-06 +9.73e
    -06 PASS
u=0.7 v=0.0656 pred=0.49765 opposite=0.53908 gaps by block 20,40,80,160: +9.93e-06 +9.93e-06 +9.93e-06 +9.93e
    -06 PASS
u=0.7 v=0.0779 pred=0.49936 opposite=0.54877 gaps by block 20,40,80,160: +3.87e-06 +3.87e-06 +3.87e-06 +3.87e
    -06 PASS
VERDICT: KILLED at the stated bar
OPPOSITE (Hermitian-form guess must fail): fails as required
```

B.9 `incoming_exit_filter.py` (imported audit, prepared with OpenAI tools)

```
"""Independent endpoint-filter audit of preserved incoming matter-disk scripts."""
from pathlib import Path
import json, math, hashlib
import numpy as np
from scipy.linalg import eigvals
```

```
ROOT=Path(__file__).resolve().parent
N=2000;L=40;t1=.5
k=2*np.pi*(np.arange(N)+.5)/N
lags=np.arange(-(L-1),L)
ph=np.exp(1j*np.outer(lags,k))/N
disp=np.arange(L)[:,None]-np.arange(L)[None,:]+L-1

def band(m,g):
    dz=m+1j*g
    q=t1+np.exp(1j*k)
    E=np.sqrt(abs(q)**2+dz*dz+0j)
    P=np.zeros((N,2,2),complex)
    P[:,0,0]=(1-dz/E)/2;P[:,1,1]=(1+dz/E)/2
    P[:,0,1]=-q.conj()/(2*E);P[:,1,0]=-q/(2*E)
    return P,E

def spectrum(m,g):
    P,E=band(m,g)
    coeff=np.einsum('lk,kab->lab',ph,P,optimize=True)
    c=coeff[disp].transpose(0,2,1,3).reshape(2*L,2*L)
    return eigvals(c,check_finite=False)

def rotation(m,g,cut):
    nu=spectrum(m,g)
    nu=nu[(np.abs(nu)>cut)&(np.abs(1-nu)>cut)]
    return float(np.max(np.abs(np.log((1-nu)/nu).imag))/np.pi) if len(nu) else 0.

def exit_at(m,cut):
    lo=0.;g=.05
    while g<.8:
        if rotation(m,g,cut)>=.5:
            hi=g;break
        lo=g;g*=1.4
    else:
        return None
    for _ in range(26):
        mid=(lo+hi)/2
        if rotation(m,mid,cut)<.5:lo=mid
        else:hi=mid
    g=(lo+hi)/2
    _,E=band(m,g)
    dnorm=(t1+np.cos(k))**2+np.sin(k)**2+abs(m+1j*g)**2
    K=float(np.mean((1+dnorm/abs(E)**2)/2))
    return dict(m=m,cut=cut,exit_gamma=g,mean_Petermann=K)

# Independent eigensolver cross-check for projector selection.
errors=[]
for m,g in ((.4,.1),(.4,.2),(0.,.4)):
    P,E=band(m,g)
    for ix in (0,137,999,1567):
        q=t1+np.exp(1j*k[ix]);h=np.array([[m+1j*g,q.conjugate()],[q,-m-1j*g]])
        e,R=np.linalg.eig(h);order=np.argsort(e.real);R=R[:,order]
        inverse=np.linalg.inv(R)
        errors.append(float(np.max(abs(P[ix]-np.outer(R[:,0],inverse[0,:])))))
assert max(errors)<1e-12
rows=[]
for m in (.4,0.):
    for cut in (1e-6,1e-8,1e-9,1e-10,1e-12):
        row=exit_at(m,cut);rows.append(row);print(row,flush=True)
oblique=np.array([[2.,1.],[-2.,-1.]])
assert np.array_equal(oblique@oblique,oblique)
report={'classification':'Endpoint-filter␣sensitivity,␣not␣a␣new␣physical␣measurement',
        'N':N,'region_cells':L,'t1':t1,'max_projector_crosscheck_error':max(errors),
        'rows':rows,'oblique_projector':{'C':oblique.tolist(),'X_A':-2.,'Gamma_b':3.},
        'protocol_sha256':hashlib.sha256((ROOT/'INCOMING_EXIT_FILTER_PROTOCOL.md').read_bytes()).hexdigest()}
(ROOT/'incoming_exit_filter_results.json').write_text(json.dumps(report,indent=2)+'\n')
```

```
{'m': 0.4, 'cut': 1e-06, 'exit_gamma': 0.2716369288976192, 'mean_Petermann': 1.0579728462043967}
{'m': 0.4, 'cut': 1e-08, 'exit_gamma': 0.2716369288976192, 'mean_Petermann': 1.0579728462043967}
{'m': 0.4, 'cut': 1e-09, 'exit_gamma': 0.16916993247628215, 'mean_Petermann': 1.0223524908929857}
{'m': 0.4, 'cut': 1e-10, 'exit_gamma': 0.16916993247628215, 'mean_Petermann': 1.0223524908929857}
{'m': 0.4, 'cut': 1e-12, 'exit_gamma': 0.1227913656055927, 'mean_Petermann': 1.0117452195516827}
{'m': 0.0, 'cut': 1e-06, 'exit_gamma': 0.4743416490950296, 'mean_Petermann': 2.000000001634376}
{'m': 0.0, 'cut': 1e-08, 'exit_gamma': 0.4743416490950296, 'mean_Petermann': 2.000000001634376}
{'m': 0.0, 'cut': 1e-09, 'exit_gamma': 0.4743416490950296, 'mean_Petermann': 2.000000001634376}
{'m': 0.0, 'cut': 1e-10, 'exit_gamma': 0.4743414673329256, 'mean_Petermann': 1.9999957440096103}
{'m': 0.0, 'cut': 1e-12, 'exit_gamma': 0.47422795577696786, 'mean_Petermann': 1.9973451918175165}
```